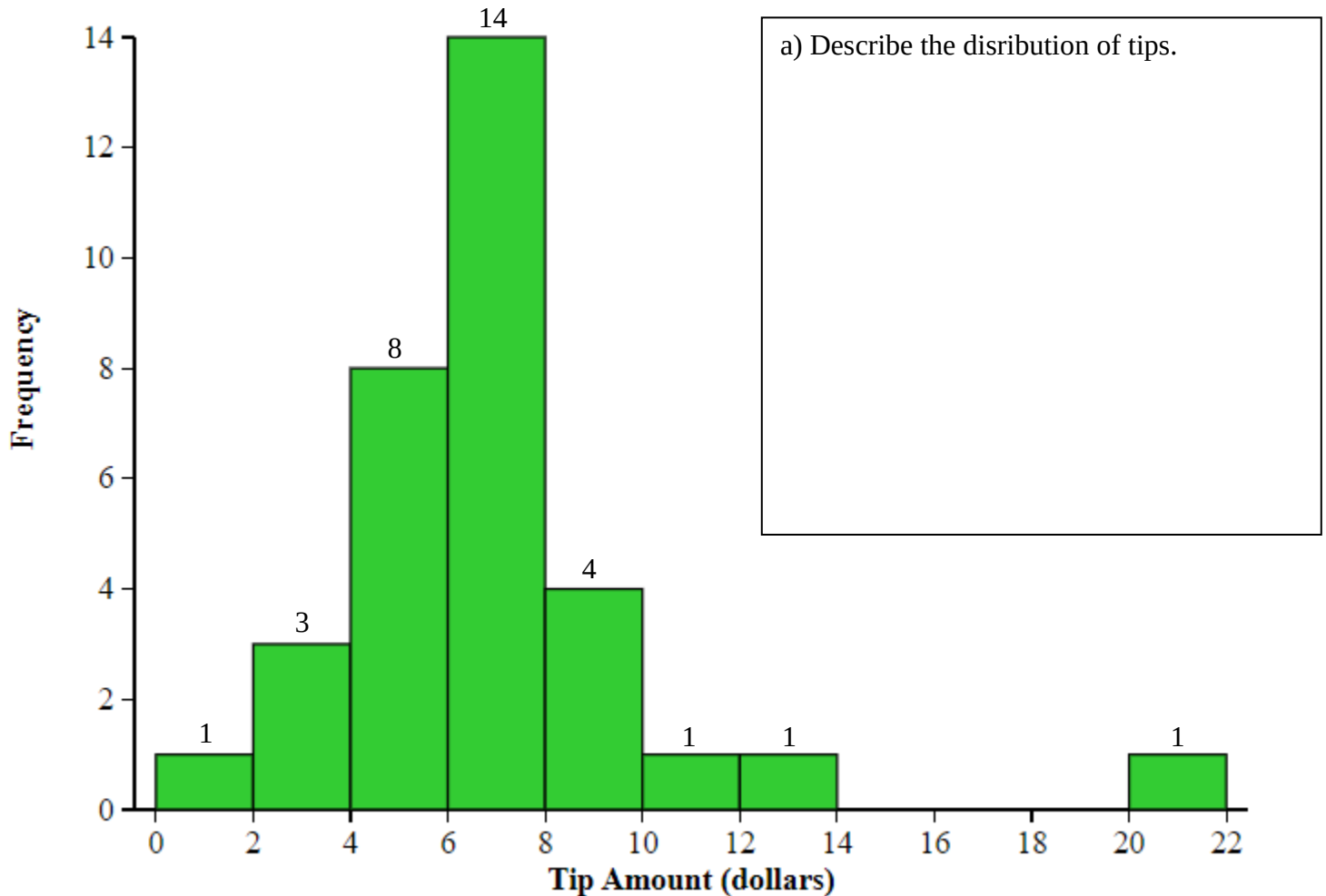


1) Karley works as a Dominoes delivery driver and on Friday night, she recorded how many tips she got from each of her deliveries. The histogram below shows the distribution of the 33 deliveries.



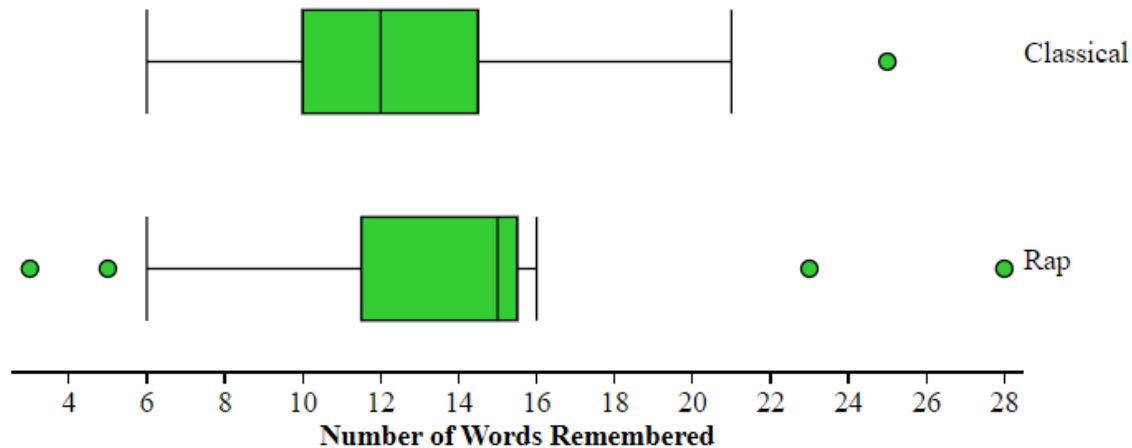
b) One of the tip amounts was \$9. If this tip had been \$14 instead of \$9, describe what effect this would have on the following measures of center. Justify your answer.

Mean:

Median:

c) Describe how you would estimate the median of this distribution.

2) Jordan and Alex want to know if listening to different types of music while studying will help you remember the material better. They randomly assigned a group of students to two different groups: one group studied a list of words while listening to classical music while the second group studied the same list of words while listening to rap music. There was a total of 30 words on the list and each group studied the list while listening to the music for 5 minutes. They were then asked to write down as many words as they could remember. The data is displayed in the boxplots below.

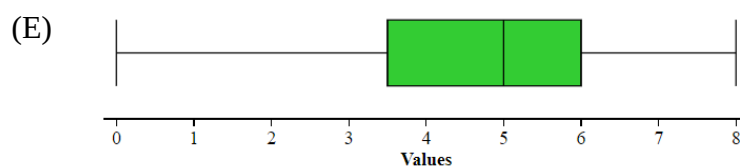
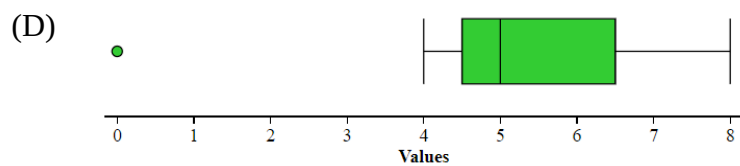
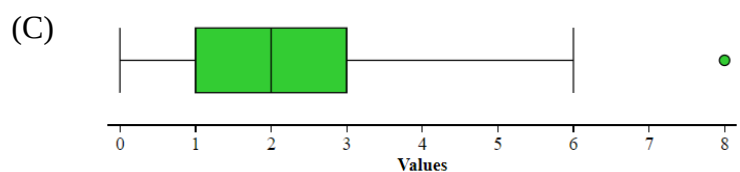
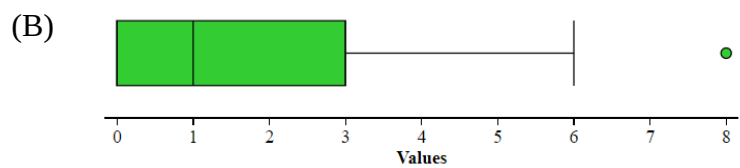
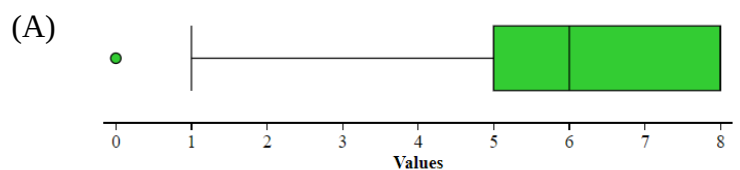
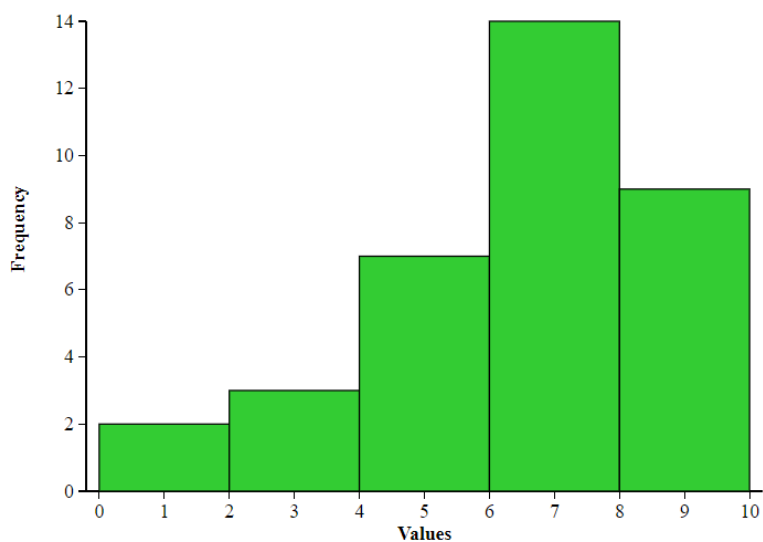


a) Approximate the interquartile range for each set of data. Why is this the appropriate measure of spread to use for these two data sets?

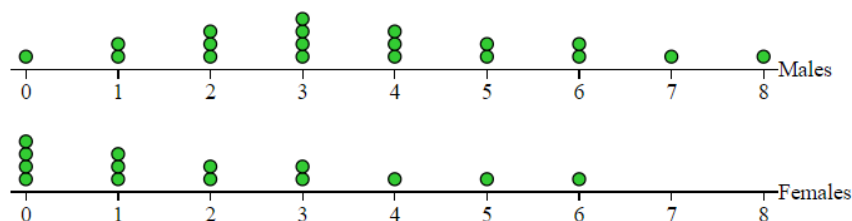
b) Write three sentences comparing the number of words remembered between each of the two groups.

Multiple Choice Practice

_____ 3) The histogram below displays a set of measurements. Which of the boxplots below displays the same set of measurements?



_____ 4) These dotplots for randomly selected male and female students at a particular high school show the number of times per week they eat at fast food restaurants.



Which of the following is a true statement?

- (A) One distribution is roughly symmetric; the other is skewed left.
- (B) The males' median is larger than the females' mean.
- (C) The ranges are both 8.
- (D) The standard deviations are equal.
- (E) Combining the male and female times into one set of student times will increase the range to 16.

_____ 5) The boxplots below summarize two sets of data, A and B. Which of the following must be true?



- I. Set A contains more data than Set B.
- II. Set A has a larger range than Set B.
- III. Set A and Set B have the same median.

- (A) I only
- (B) III only
- (C) I and II only
- (D) II and III only
- (E) I, II, and III

_____ 6) Which of the following statements is incorrect?

- (A) Both dotplots and stemplots can show symmetry, gaps, clusters, and outliers.
- (B) Sets with different distribution shapes can have identical boxplots.
- (C) Boxplots, dotplots, stemplots, and histograms can all show skewness.
- (D) In histograms, relative areas correspond to relative frequencies.
- (E) In histograms, frequencies can be determined from relative heights.

_____ 7) Consider the following test scores:

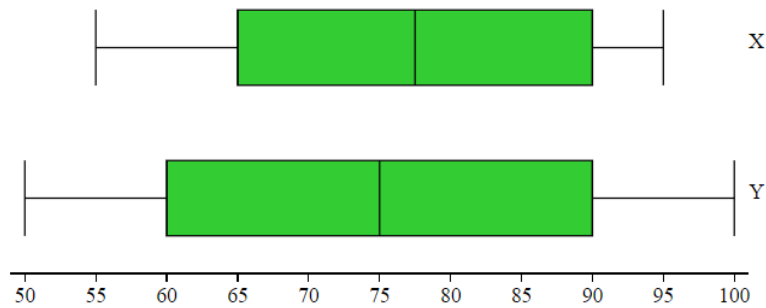
Class A: 65, 72, 73, 73, 74, 82, 91

Class B: 60, 65, 79, 81, 85, 86, 86

Which of the following is a true statement?

- (A) Class A has the greater range, while Class B has the greater standard deviation.
- (B) Class B has the greater range, while Class A has the greater standard deviation.
- (C) The ranges and standard deviations are the same for both classes.
- (D) The ranges are the same, but Class A has the higher standard deviation.
- (E) The ranges are the same, but Class B has the higher standard deviation.

_____ 8) The boxplots shown below summarize two sets of data, X and Y.



Which of the following must be true?

- (A) 50 is an outlier for Set Y.
- (B) Set Y contains more data than Set X.
- (C) The IQR of Set X is smaller than the IQR of Set Y.
- (D) The range of Set Y is smaller than the range of Set X.
- (E) The means of the two data sets are equal.